

Perfect Secrecy

For a cryptosystem (M, C, K, E, D) where:

$x \in M$, the set of all plaintext strings

$y \in C$, the set of all ciphertext strings

$k \in K$, the set of all encryption keys

$e_k \in E$, the set of all encryption rules for a key k (every rule can be the same encryption function)

$d_k \in D$, the set of all decryption rules for a key k (every rule can be the same decryption function)

This cryptosystem has perfect secrecy if $P(x | y) = P(x)$ for all $x \in M$ and $y \in C$.

Example:

A cryptosystem consists of the following plaintext, ciphertext, and key sets:

$x \in M = \{a, b\}$, the set of all plaintext strings

$y \in C = \{1, 2, 3, 4\}$, the set of all ciphertext strings

$k \in K = \{k_1, k_2, k_3\}$, the set of all encryption keys

The probability distributions for x and k are: $P(a) = \frac{1}{4}, P(b) = \frac{3}{4}$ and $P(k_1) = \frac{1}{2}, P(k_2) = \frac{1}{4}, P(k_3) = \frac{1}{4}$

The encryption matrix is:

	a	b
k_1	1	2
k_2	2	3
k_3	3	4

Is the perfect secrecy property satisfied for any $y \in C$?

Computing $P(x = a | y = 3)$ and $P(x = b | y = 3)$, we find that the perfect secrecy property is satisfied for $y = 3$:

$$P(x = a | y = 3) = \frac{P(x = a) \cdot P(y = 3 | x = a)}{P(y = 3)} = \frac{\frac{1}{4} \cdot \frac{1}{4}}{\frac{1}{4} \times \frac{3}{4} + \frac{1}{4} \times \frac{1}{4}} = \frac{\frac{1}{16}}{\frac{3}{16} + \frac{1}{16}} = \frac{1}{16} \times \frac{16}{4} = \frac{1}{4}, \text{ and } P(a) = \frac{1}{4}$$

$$P(x = b | y = 3) = \frac{P(x = b) \cdot P(y = 3 | x = b)}{P(y = 3)} = \frac{\frac{3}{4} \cdot \frac{1}{4}}{\frac{1}{4} \times \frac{3}{4} + \frac{1}{4} \times \frac{1}{4}} = \frac{\frac{3}{16}}{\frac{3}{16} + \frac{1}{16}} = \frac{3}{16} \times \frac{16}{4} = \frac{3}{4}, \text{ and } P(b) = \frac{3}{4}$$

This cryptosystem does NOT have perfect secrecy, as the perfect secrecy property is NOT satisfied for $y \in \{1, 2, 4\}$ (see the exercises).

Exercises:

1. A cryptosystem has encryption matrix:

	a	b
k_1	1	2
k_2	2	3
k_3	3	4

And probability distributions: $P(a) = \frac{1}{4}, P(b) = \frac{3}{4}$ and $P(k_1) = \frac{1}{2}, P(k_2) = \frac{1}{4}, P(k_3) = \frac{1}{4}$

Prove that this cryptosystem does NOT have perfect secrecy by showing that:

- a) $P(x | y = 1) \neq P(x)$ for all $x \in M$.
- b) $P(x | y = 2) \neq P(x)$ for all $x \in M$.
- c) $P(x | y = 4) \neq P(x)$ for all $x \in M$.

2. A cryptosystem consists of the following plaintext, ciphertext, and key sets:

$x \in M = \{0,1\}$, the set of all plaintext strings

$y \in C = \{0,1\}$, the set of all ciphertext strings

$k \in K = \{k_1, k_2\}$, the set of all encryption keys

The probability distributions for x and k are:

$$P(x=0) = \frac{1}{3}, P(x=1) = \frac{2}{3} \text{ and } P(k_1) = \frac{1}{2}, P(k_2) = \frac{1}{2}$$

The encryption matrix is:

	0	1
k_1	1	0
k_2	0	1

- a) Compute $P(x | y = 0)$ for all $x \in M$.
- b) Compute $P(x | y = 1)$ for all $x \in M$.
- c) Does this cryptosystem have perfect secrecy?

Answers:

1.

a) $P(x = a | y = 1) = 1 \neq \frac{1}{4}$, $P(x = b | y = 1) = 0 \neq \frac{3}{4}$

b) $P(x = a | y = 2) = \frac{1}{7} \neq \frac{1}{4}$, $P(x = b | y = 2) = \frac{6}{7} \neq \frac{3}{4}$

c) $P(x = a | y = 4) = 0 \neq \frac{1}{4}$, $P(x = b | y = 4) = 1 \neq \frac{3}{4}$

2.

a) $P(x = 0 | y = 0) = \frac{1}{3}$, $P(x = 1 | y = 0) = \frac{2}{3}$

b) $P(x = 0 | y = 1) = \frac{1}{3}$, $P(x = 1 | y = 1) = \frac{2}{3}$

c) Yes, $P(x | y) = P(x)$ for all $x \in M$ and $y \in C$.